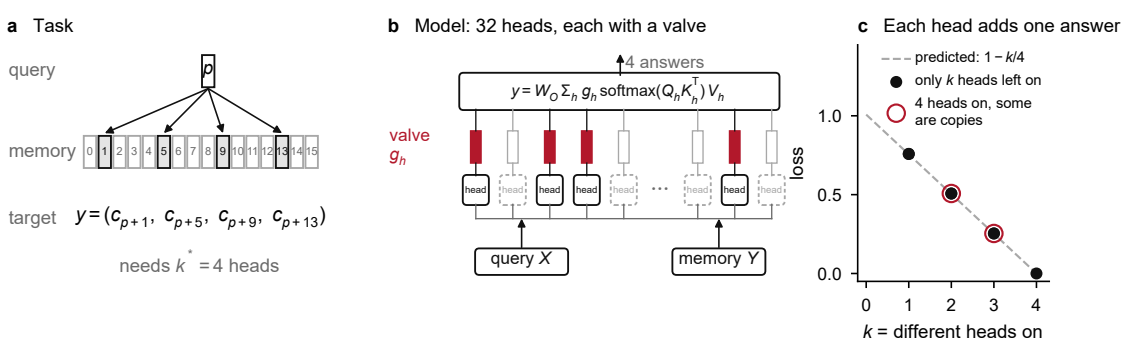

a Task
query
p
memory
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
target
$y = (c_{p+1}, c_{p+5}, c_{p+9}, c_{p+13})$
needs $k^* = 4$ heads
b Model: 32 heads, each with a valve
4 answers
$y = W_O \Sigma_h g_h \text{softmax}(Q_h K_h^T) V_h$
valve g_h
head
query X
memory Y
c Each head adds one answer
predicted: $1 - k/4$
only k heads left on
4 heads on, some are copies
loss
1.0
0.5
0.0
0 1 2 3 4
k = different heads on